

Jain Koo

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RESEARCH INTEREST

- **Computer Architecture and Systems for Efficient Machine Learning**
- **Hardware–Software Co-design for ML Accelerator System**
- **Domain-Specific Accelerators**

EDUCATION

Master of Science <i>Electrical and Electronic Engineering</i> Yonsei University (Advisor : Prof. Won Woo Ro)	Sep 2025 – Aug 2027 Seoul, Korea
Bachelor of Science <i>Electrical and Electronic Engineering, Earth System Science</i> Yonsei University	Mar. 2021 – Aug 2025 Seoul, Korea

PUBLICATIONS

- [1] **Jain Koo**, Sungbin Kim, Enhyeok Jang, Sungwoo Kim, Yingyan (Celine) Lin, and Won Woo Ro, “**DiVA: Algorithm–System Co-Design for Visual Autoregressive Models via Submap-Level Retrievals**,” *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, In Submission, 2027
- [2] Hoje Lee, Minchan Kim, Jaewon Kwon, **Jain Koo**, Yongju Lee, Won Hur, Jiwon Lee, and Won Woo Ro, “**Maverick: Leveraging Address Translation to Mitigate Memory Latency in Server Workloads**,” *Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2026.
- [3] **Jain Koo**, Enhyeok Jang, and Won Woo Ro, “Key–Value Cache Retrieval Is No Longer Cheap in Visual Autoregressive Models,” *The 41st International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC)*, 2026.

PROJECTS AND RESEARCH

Algorithm–System Co-design for Efficient Image Generation • Emerging VAR workload characterization and bottleneck profiling • Build an efficient hardware system for KV cache retrieval	Jan 2025 – Present
Memory-Centric Architecture Using Reconfigurable PIM Devices • Accelerating attention computation with multiple PIM devices • Exploiting pipeline & tensor parallelism for latency and throughput enhancement • Evaluated inter-PIM network overhead with BookSim2.0 simulator	Sep 2025 – Present
Development of PIM Software Architecture based on Data-Flow Computing • Developed a dynamic resource-aware tiling framework for Multi-PIM platform • Jointly optimized workload partitioning and PIM resource allocation	Jan 2025 – Aug 2025
In-DRAM PIM-Based On-Device AI Acceleration • Funded by Samsung Electronics • Bottleneck profiling of Language Models (LMs) workload • Compute offloading to PIM in GPU-PIM heterogeneous computing platform	Jan 2025 – Aug 2025

RESEARCH EXPERIENCE

Graduate Student Researcher

Sep 2025 – Aug 2027

Embedded Systems and Computer Architecture Lab (eSCaL)

Yonsei University

- Research on hardware–software co-design for efficient ML systems.
- Focused on ML acceleration in heterogeneous computing systems.
- Focused on data movement, execution flow and communication optimization.

Undergraduate Student Researcher

Jan 2025 – Sep 2025

Embedded Systems and Computer Architecture Lab (eSCaL)

Yonsei University

- Focused on bottleneck profiling and explored hardware–software optimization techniques.
- Focused on hardware-aware efficient ML algorithms and systems.

Undergraduate Student Researcher

Jan 2024 – Sep 2024

Multidimensional Insight Lab (Advisor : Prof. Sanghoon Lee)

Yonsei University

- Focused on efficient computer vision systems.
- Focused on hardware-aware efficient computer vision algorithms.

SCHOLARSHIPS AND AWARDS

BK (Brain Korea) 21 Research Fellowship

2025 – 2027

Graduate research fellowship funded through the BK21 program

Yonsei University

J.H. Foundation Research Fellowship

2026 – 2027

Merit-based scholarship for outstanding academic performance

J.H. Cultural Foundation

Graduate Student Research Assistant (GSRA) Scholarship

2025

Graduate research scholarship supporting research activities

Yonsei University

Academic Honors

2024

Recognition for outstanding academic performance

Yonsei University

The Yonsei Jinri Scholarship

2024

Merit-based scholarship for outstanding academic performance

Yonsei University

TEACHING EXPERIENCE

Electrical and Electronic Engineering Capstone Design (EEE4610) TA

Spring 2026

- Instructing groups of undergraduate students working on the Bachelors dissertation
- Directing research projects focused on accelerating Large Language Models and memory systems
- Hosting weekly meeting and advising the students on their research directions

CONTRIBUTED TALKS

Efficient KV Cache Retrieval for Generative AI Models

May, 2026

Computer Architecture for Efficient ML Systems Seminar

Seoul, Korea

Efficient Large Language Model Systems

Jan, 2026

Efficient Large Language Model (LLM) Systems Seminar

Seoul, Korea

Understanding GPU Microarchitecture

Aug, 2025

Advanced Computer Architecture Seminar

Seoul, Korea

SKILLS

Languages: Korean (Native), English (CEFR C1, TOEFL iBT 110/120)

Programming: Python, C, C++, Verilog, CUDA, PyTorch